01 主功率级 / 15–25V → 10V, 1A

Rev B / Native editable draft / 分流器Kelvin取样；MOS漏极背板分别绝缘；模拟电阻负载；机械封装待确认。

J1



2P POWER

TERM_5.08_2P [TBD]

J2



2P POWER

TERM_5.08_2P [TBD]

F1



2A / 5x20

FUSE_5X20_P22.5_TBD [TBD]

D1



SS56

SMB_DO214AA

RS_IN



50mΩ / 1% / >=1W

R2512

RS_OUT



50mΩ / 1% / >=1W

R2512

RS_L



25mΩ / 1% / >=1W

R2512

C1



470uF / 50V

CP_D12.5_P5_TBD [TBD]

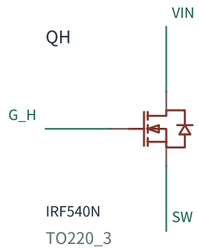
C2



1uF / 50V

C1206

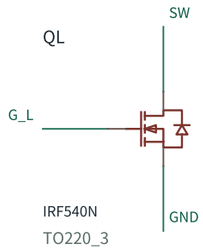
QH



IRF540N

TO220_3

QL



IRF540N

TO220_3

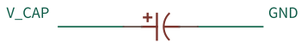
L1



220uH / Isat>=3A

L_TH_D18_P10_TBD [TBD]

C3



470uF / 35V

CP_D12.5_P5_TBD [TBD]

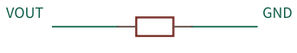
C4



10uF / 35V

C1206

R_BL



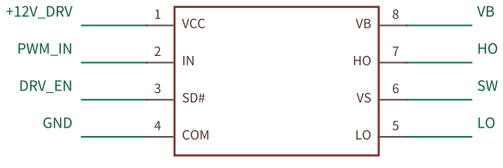
2.2kΩ / 0.25W

R1206

02 IR2104驱动与辅助电源接口

Rev B / Native editable draft / DRV_EN=0关闭两路；IN=0不能代替停机；辅助电源为外接模块。

U1



IR2104PbF

DIP8_300

D_BOOT



UF4007

DO41_P10.16 [TBD]

RG_H



10Ω

R0805

RG_L



10Ω

R0805

R_GSH



10kΩ

R0805

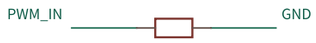
R_GSL



10kΩ

R0805

R_IN_PD



10kΩ

R0805

R_SD_PD



100kΩ

R0805

R_PWM



100Ω

R0805

C_BOOT



1uF / 25V

C0805

C5



100nF

C0805

C6



10uF / 25V

C1206

J_AUX12



VIN→12V >=0.2A

HDR_1X4_2.54 [TBD]

J_AUX5



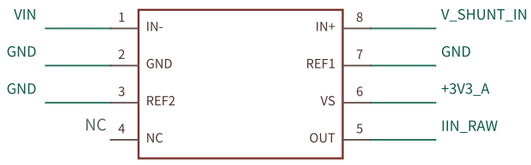
VIN→5V >=0.5A

HDR_1X4_2.54 [TBD]

03 电流采样与模拟电源

Rev B / Native editable draft / INA240A1D使用SOIC-8脚序；U4零点=3V3_A/2，增益0.5V/A；NC脚悬空。

U2



INA240A1D

SOIC8_150

U4



INA240A1D

SOIC8_150

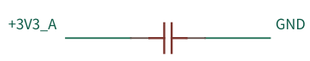
U3



INA181A1DBV

SOT23_6

C_U2



100nF

C0805

C_U3



100nF

C0805

C_U4



100nF

C0805

C_A2



100nF

C0805

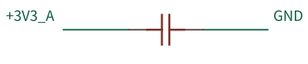
R_VDDA



10Ω

R0805

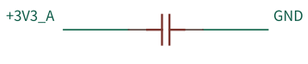
C_A1



1uF

C0805

C_A3

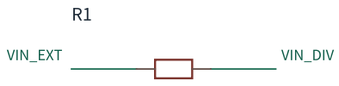


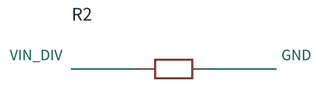
10uF / 25V

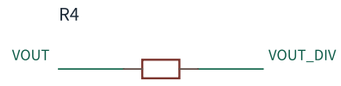
C1206

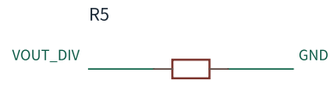
04 电压分压、五路ADC滤波与钳位

Rev B / Native editable draft / 两路电压分压均为11:1；比较器使用原始IL_BI/VOUT_DIV；BAT54S不是BAT54。







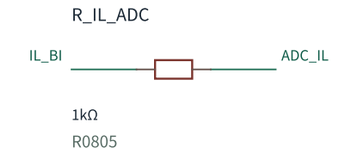




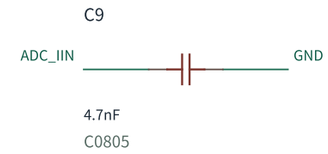






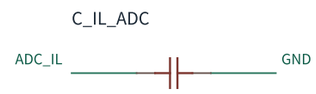


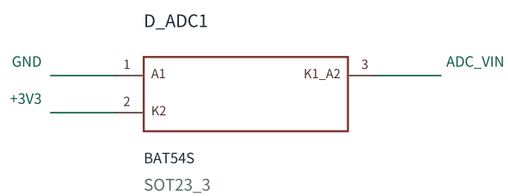


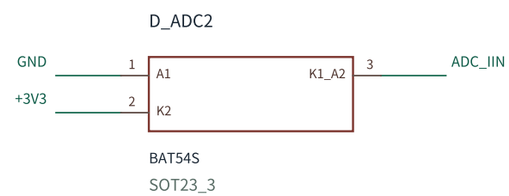


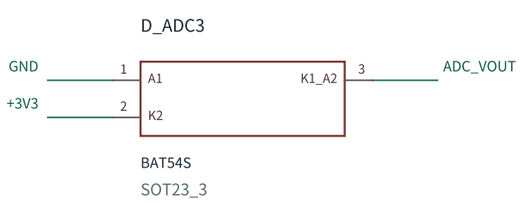


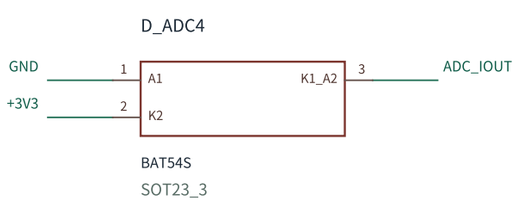


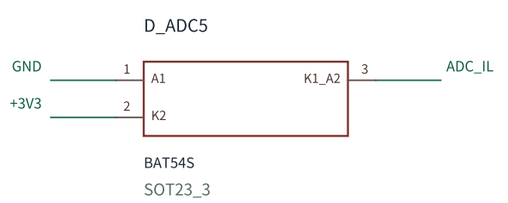












05 正负过流及过压比较保护

Rev B / Native editable draft / FAULT_N开集电极汇合，仅上拉至3.3V；门限标称±1.8A/11V，需校准。

U5



LM393

SOIC8_150

U9



LM393

SOIC8_150

R9



10kΩ / 0.1%

R0805

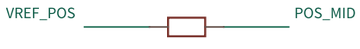
R16



10kΩ / 0.1%

R0805

R10A



33kΩ / 0.1%

R0805

R15A



33kΩ / 0.1%

R0805

R10B



1kΩ / 0.1%

R0805

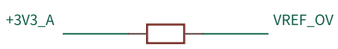
R15B



1kΩ / 0.1%

R0805

R11



23kΩ

R0805

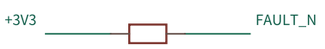
R12



10kΩ

R0805

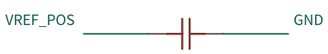
R13



10kΩ

R0805

C_REF_POS



100nF

C0805

C_REF_NEG



100nF

C0805

C_REF_OV



100nF

C0805

C_U5



100nF

C0805

C_U9



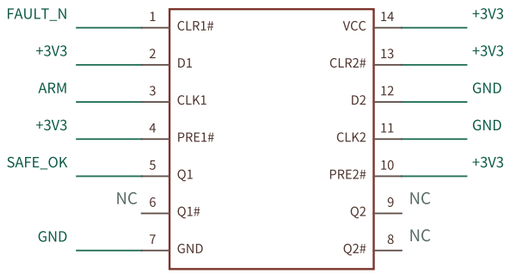
100nF

C0805

06 上电复位、故障锁存与使能

Rev B / Native editable draft / SAFE_OK AND MCU_EN → DRV_EN；故障清除后必须重新ARM；未使用输出悬空。

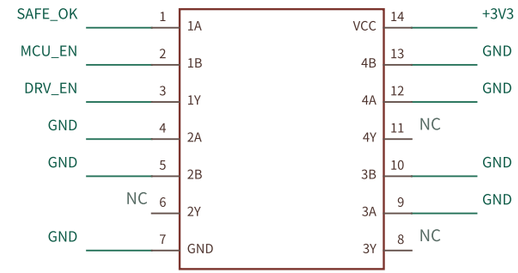
U6



SN74HC74DR

SOIC14_150

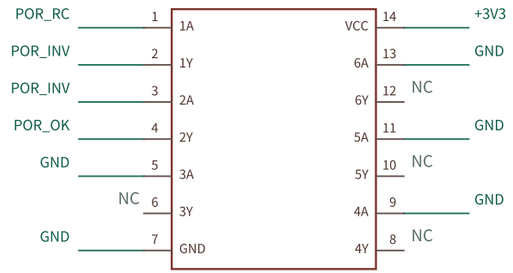
U7



SN74HC08DR

SOIC14_150

U8



SN74HC14DR

SOIC14_150

R14



10kΩ

R0805

R_EN_PD



10kΩ

R0805

R_ARM_PD



10kΩ

R0805

C11



100nF

C0805

C_U6



100nF

C0805

C_U7



100nF

C0805

C_U8



100nF

C0805

D_POR



BAT54

SOT23_3

07 STM32控制板接口与测试点

Rev B / Native editable draft / LCD/按键/STM32核心板在外接板；3.3V由核心板回送；避免USB形成双电源。

